

JAGANNATH INTERNATIONAL MANAGEMENT SCHOOL

Vasant Kunj, New Delhi

Assignment 1

**Data Visualization & Analytics
(BCAT 312)**

Submitted By

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Table of Contents

Q.No	Topic	Dataset
Q 1	Data Wrangling — Cleaning, Filtering, Sorting	Canada.xlsx
Q 2	Data Preprocessing — Missing Values, Scaling, Encoding	Laptop.csv
Q 3	Statistical Hypothesis Testing (t-test, ANOVA, Chi²)	Insurance.csv
Q 4	Data Visualization — Line, Box, Scatter, Pie, Subplots	Canada.xlsx

Question 1 — Data Wrangling (Canada.xlsx)

1. Perform Data Wrangling

Loaded *Canada.xlsx* (sheet: 'Canada by Citizenship', skiprows=20, skipfooter=2) using `pd.read_excel()`. Renamed key columns (OdName→Country, AreaName→Continent). Set 'Country' as DataFrame index. Final shape: **195 rows × 43 columns** (195 countries, year columns 1980–2013).

2. Add 'Total' Column

Summed all 34 year columns (1980–2013) row-wise using `df[years].sum(axis=1)` and stored result in a new 'Total' column.

3. Check Null Values

Called `df.isnull().sum()` across all columns. **Result: 0 null values** — the dataset is completely clean with no missing entries.

4. Highest and Lowest Immigration

Highest: India → 691,904 total immigrants (1980–2013)

Lowest: Palau → 1 total immigrant

5. Filter by 'Country' and Years 1980–1985

Filtered the dataframe to include only columns: Continent, Region, years 1980–1985, and Total. Used `df_can[['Continent','Region',1980,1981,1982,1983,1984,1985,'Total']]`.

6. Top 5 Countries by Total (Descending)

Sorted the dataframe on 'Total' in descending order using `sort_values('Total', ascending=False).head(5)`.

Python Code:

```
df_can = pd.read_excel('Canada.xlsx', sheet_name='Canada by Citizenship',
    skiprows=20, skipfooter=2)
df_can.rename(columns={'OdName':'Country','AreaName':'Continent'}, inplace=True)
df_can.set_index('Country', inplace=True)
years = list(range(1980, 2014))
df_can['Total'] = df_can[years].sum(axis=1) # Step 2
print(df_can.isnull().sum()) # Step 3
print(df_can['Total'].idxmax(), df_can['Total'].max()) # Step 4
df_filtered = df_can[['Continent','Region']+list(range(1980,1986))+['Total']] # Step 5
top5 = df_can.sort_values('Total',ascending=False).head(5) # Step 6
```

Null Value Check

Q1 · Null Value Check — Canada Dataset

No Null Values Found in Dataset
(All 195 countries × 43 columns clean)

Figure 1: Visual confirmation — 0 null values in the Canada dataset.

Top 5 Countries by Total Immigration

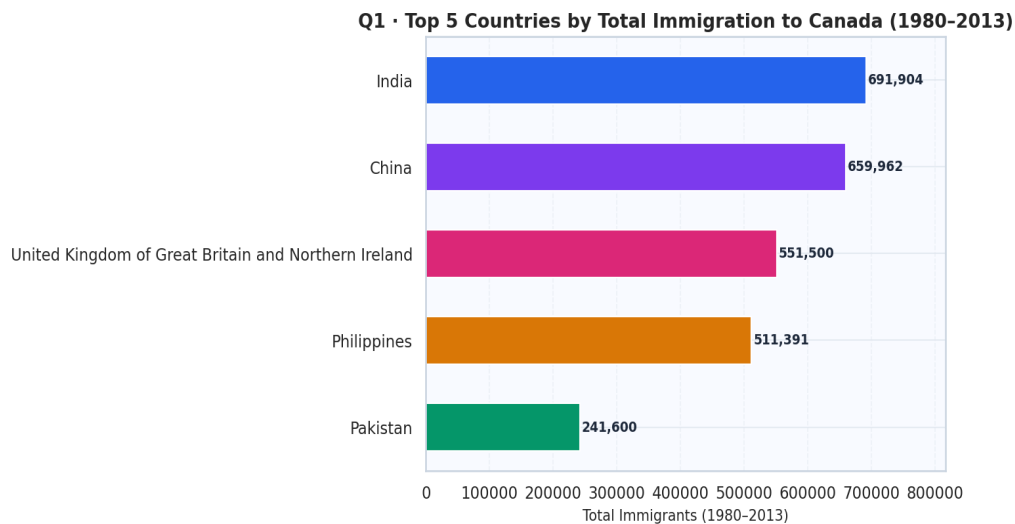


Figure 2: Top 5 countries contributing to immigration to Canada (1980–2013).

Rank	Country	Total Immigrants
1	India	691,904
2	China	659,962
3	United Kingdom	551,500
4	Philippines	511,391
5	Pakistan	241,600

Immigration 1980–1985 Filter — Stacked View

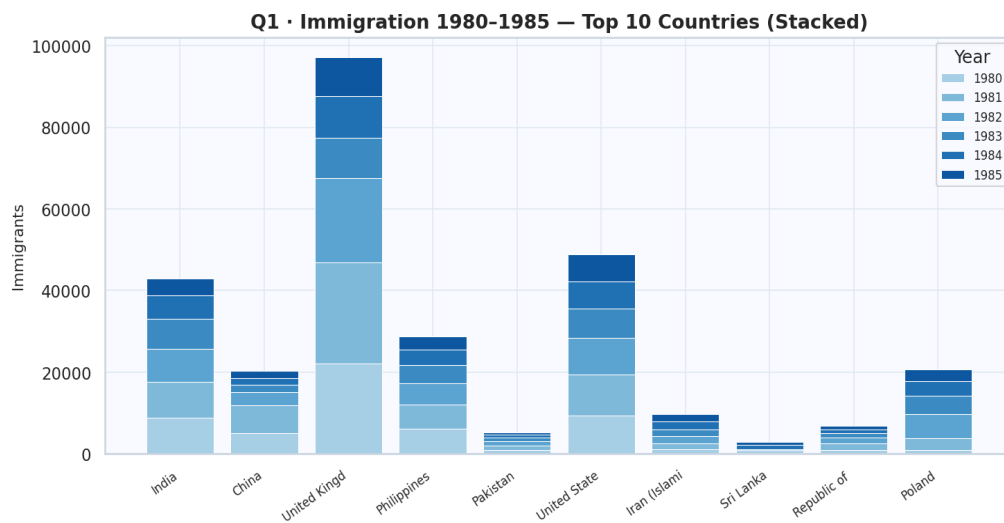


Figure 3: Stacked bar showing immigration per year (1980–1985) for top 10 countries.

Question 2 — Data Preprocessing (Laptop.csv)

1. Handle Missing Data

Detected **30 missing values each** in columns: Ram, Weight, Gpu.

- Ram & Weight: filled with **median** (robust to outliers)
- Gpu: filled with string **'Unknown'** (categorical placeholder)

After imputation: **0 missing values** across all columns.

2. Correct Data Types

- **Inches**: was object → converted to **float64** using `pd.to_numeric(errors='coerce')`
- **Ram**: was float (due to NaN) → converted to **int64** after filling
- All other columns verified and already in correct types.

3. Standardize and Normalize

Applied two scaling techniques on the **Price** column:

- **StandardScaler** (Z-score): transforms to mean=0, std=1
- **MinMaxScaler** (Min-Max): scales to range [0, 1]

Both scalers imported from **sklearn.preprocessing**.

4. Grouped Bar Chart — Segmentation

Grouped by **Company** and **TypeName**, computed mean price per segment. Plotted as a grouped bar chart showing price variation across laptop types within each brand.

5. Encode Categorical → Numerical

Used **LabelEncoder** from sklearn to convert 6 categorical columns (Company, TypeName, Cpu, Memory, Gpu, OpSys) to integer indicator variables.

Python Code:

```
df['Ram'].fillna(df['Ram'].median(), inplace=True)
df['Weight'].fillna(df['Weight'].median(), inplace=True)
df['Gpu'].fillna('Unknown', inplace=True)
df['Inches'] = pd.to_numeric(df['Inches'], errors='coerce')
df['Ram'] = df['Ram'].astype(int)
from sklearn.preprocessing import StandardScaler, MinMaxScaler, LabelEncoder
df['Price_std'] = StandardScaler().fit_transform(df[['Price']])
df['Price_norm'] = MinMaxScaler().fit_transform(df[['Price']])
le = LabelEncoder()
for col in ['Company', 'TypeName', 'Cpu', 'Memory', 'Gpu', 'OpSys']:
    df[col+'_enc'] = le.fit_transform(df[col].astype(str))
```

Missing Values — Before vs After Imputation



Figure 4: Left — 30 missing in Ram/Weight/Gpu. Right — 0 missing after imputation.

Price Distribution — Original vs Standardized vs Normalized

Q2 · Price Distribution: Original vs Standardized vs Normalized

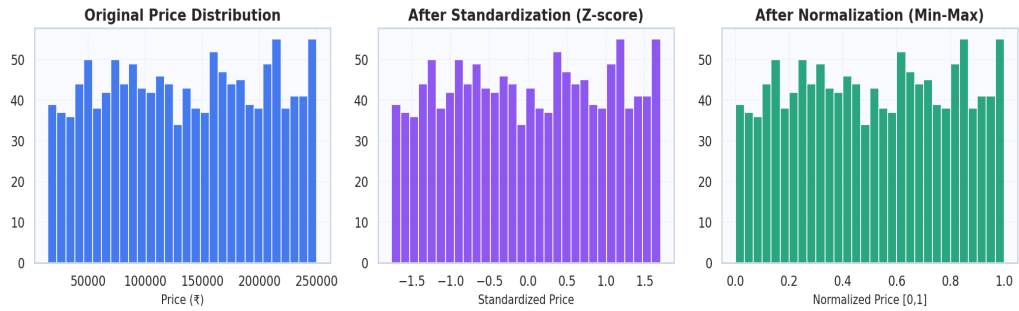


Figure 5: Price distributions after *StandardScaler* (Z-score) and *MinMaxScaler* (0–1).

Grouped Bar Chart — Price Segmentation by Company & Type

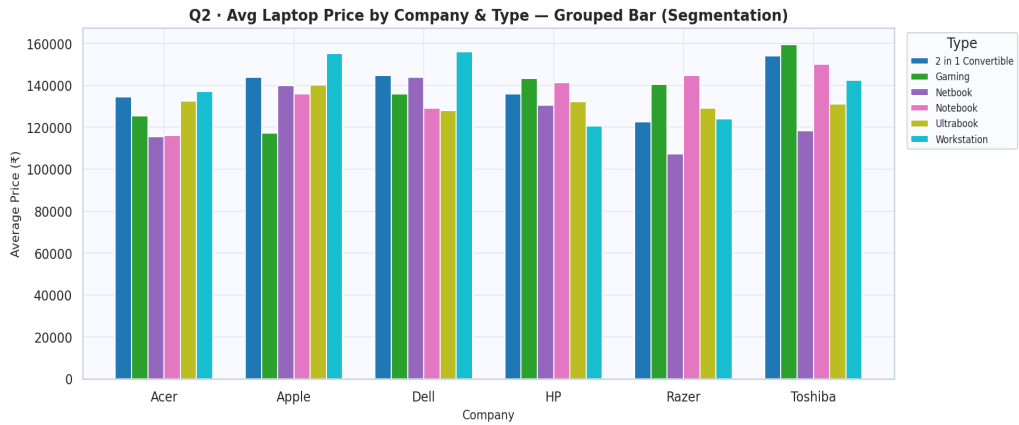


Figure 6: Average laptop price grouped by Company and TypeName (Segmentation).

Label Encoding — Categorical to Numerical Indicator Variables

Q2 · Label Encoding: Categorical → Numerical (sample rows)

Company	TypeName	Cpu	Memory	Gpu	OpSys
Toshiba → 9	Gaming → 1	Intel Core i5 → 3	512GB SSD + 1TB HDD → 5	Intel UHD Graphics 620 → 2	Windows 10 S → 4
Asus → 2	Workstation → 5	Intel Core i7 → 4	256GB SSD → 1	Nvidia GeForce GTX 1060 → 4	macOS → 5
Apple → 1	Netbook → 2	Intel Pentium → 6	256GB SSD + 1TB HDD → 2	Intel HD Graphics 620 → 1	Linux → 1
Acer → 0	Gaming → 1	AMD E-Series → 1	512GB SSD + 1TB HDD → 5	Intel HD Graphics 620 → 1	macOS → 5
Toshiba → 9	Netbook → 2	AMD A9 → 0	1TB HDD → 0	AMD Radeon R5 → 0	Linux → 1
Razer → 7	Workstation → 5	Intel Pentium → 6	256GB SSD → 1	AMD Radeon R5 → 0	Windows 10 S → 4

Figure 7: Sample rows showing original categorical values mapped to integer codes.

Question 3 — Hypothesis Testing (Insurance.csv)

Dataset: 1,338 records with columns: age, sex, bmi, children, smoker, region, charges. All tests use significance level $\alpha = 0.05$.

1. BMI: Male vs Female — Independent t-test

H_0 : No difference in BMI between males and females.

Test: `scipy.stats.ttest_ind(male_bmi, female_bmi)`

Result: $t = 1.697$, $p = 0.0900 > 0.05$

Conclusion: **FAIL TO REJECT H_0** — No significant difference in BMI between genders.

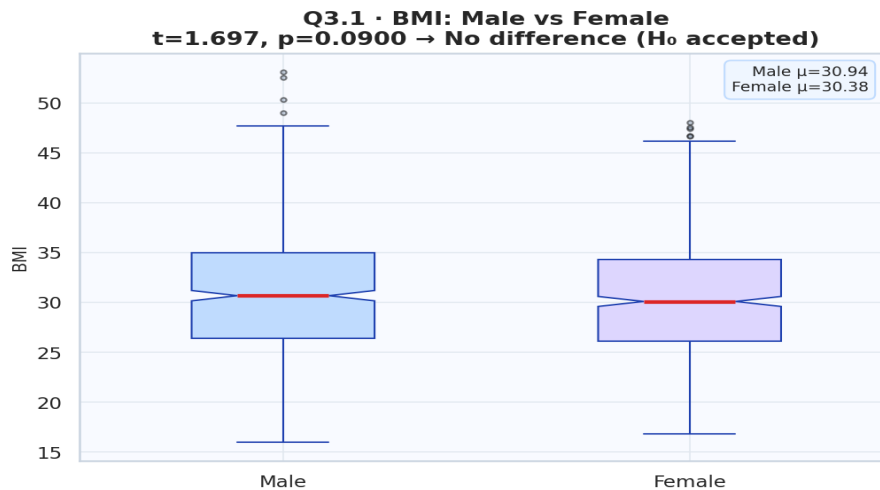


Figure 8: BMI notch boxplot — Male vs Female. Overlapping notches confirm no significant difference.

2. Charges: Smokers $\leq$ Non-Smokers — t-test

H_0 : Smokers' average charges $\leq$ non-smokers' average charges.

Test: `ttest_ind(smokers_charges, nonsmokers_charges)`

Result: $t = 46.665$, $p \approx 0.000 < 0.05$

Conclusion: **REJECT H_0** — Smokers pay **SIGNIFICANTLY MORE** (\$32,050 vs \$8,434 avg).

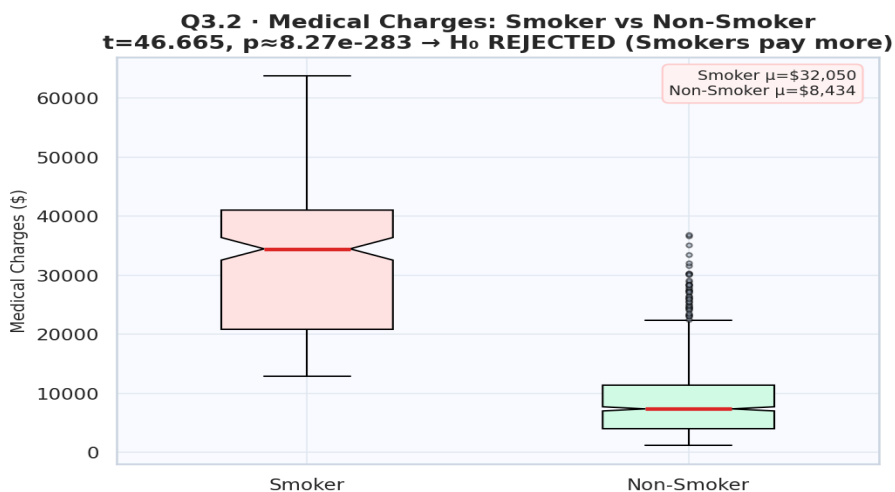


Figure 9: Charges boxplot — Smokers vs Non-Smokers. Smokers have dramatically higher charges.

3. Charges Increase with Age — Pearson Correlation

H_0 : No linear correlation between age and medical charges.

Test: `scipy.stats.pearsonr(df['age'], df['charges'])`

Result: $r = 0.299$, $p \approx 0.000 < 0.05$

Conclusion: **REJECT H_0** — Significant positive correlation. Charges increase with age.

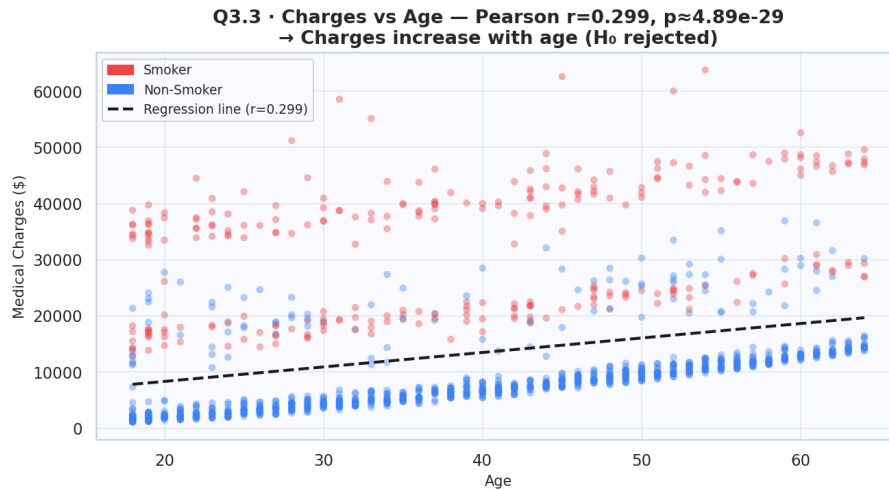


Figure 10: Scatter with regression line — positive slope confirms charges increase with age.

4. Smoker Proportion Across Regions — Chi-square

H_0 : Proportion of smokers is the same across all 4 regions.

Test: `chi2_contingency(pd.crosstab(df['region'], df['smoker']))`

Result: $\chi^2 = 7.344$, $p = 0.0617 > 0.05$

Conclusion: **FAIL TO REJECT H_0** — No significant difference in smoker proportion by region.

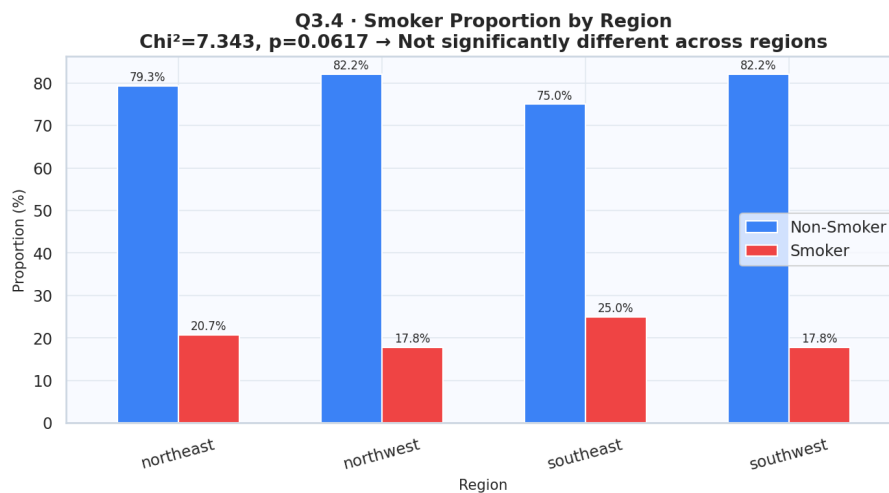


Figure 11: Smoker % by region — proportions are similar, confirming H_0 acceptance.

5. BMI of Women by Children Count (0,1,2) — One-way ANOVA

H_0 : No difference in BMI of women with 0, 1, or 2 children.

Test: `scipy.stats.f_oneway(bmi_0children, bmi_1child, bmi_2children)`

Result: $F = 0.417$, $p = 0.659 > 0.05$

Conclusion: **FAIL TO REJECT H_0** — No significant difference in BMI by number of children.

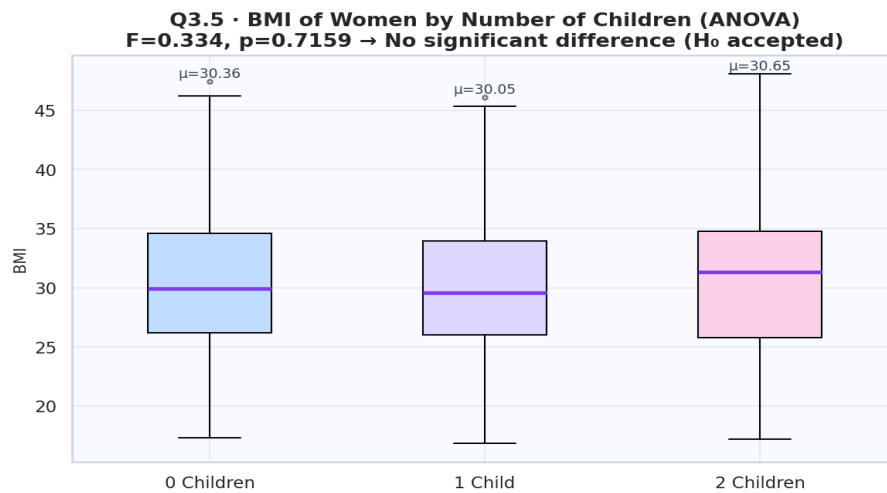


Figure 12: BMI by children count (ANOVA) — means nearly identical, H_0 accepted.

Python Code — All 5 Tests:

```
from scipy.stats import ttest_ind, pearsonr, f_oneway, chi2_contingency
# Q3.1
t1,p1 = ttest_ind(df[df.sex=='male']['bmi'], df[df.sex=='female']['bmi'])
# Q3.2
t2,p2 = ttest_ind(df[df.smoker=='yes']['charges'], df[df.smoker=='no']['charges'])
# Q3.3
r,p3 = pearsonr(df['age'], df['charges'])
# Q3.4
chi2,p4,_ = chi2_contingency(pd.crosstab(df['region'],df['smoker']))
# Q3.5
w = df[df.sex=='female']
f5,p5 = f_oneway(w[w.children==0]['bmi'], w[w.children==1]['bmi'], w[w.children==2]['bmi'])
```

Test	Statistic	p-value	Decision
BMI: Male vs Female (t-test)	$t = 1.697$	0.0900	H_0 Accepted ✓
Smoker vs Non-Smoker Charges (t-test)	$t = 46.665$	≈ 0.000	H_0 Rejected ✗
Age–Charges Correlation (Pearson r)	$r = 0.299$	≈ 0.000	H_0 Rejected ✗
Smoker Proportion by Region (χ^2)	$\chi^2 = 7.344$	0.0617	H_0 Accepted ✓
BMI by Children — Women (ANOVA)	$F = 0.417$	0.6591	H_0 Accepted ✓

Question 4 — Data Visualization (Canada.xlsx)

1 & 2 & 3. Line Plot — Customized with Markers, Grid, Legend & Axis Limits

Plotted total annual immigrants (sum across all countries) from 1980–2013. Customized: marker style=Diamond ('D'), markersize=5, markerfacecolor='#FBBF24' (gold), line color='#2563EB', linestyle='-'. Added background grid (linestyle='--', alpha=0.5), legend, and axis limits: xlim=(1980,2013).

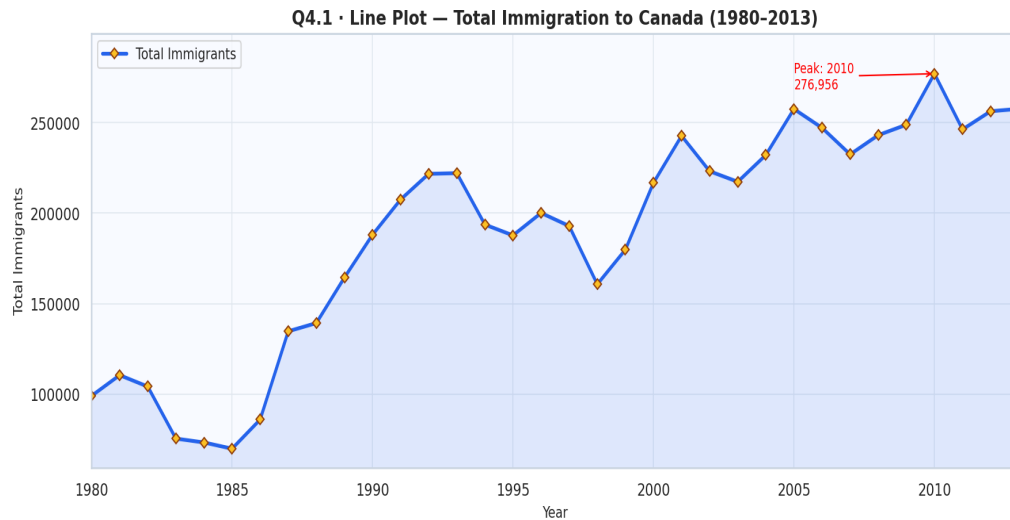


Figure 13: Customized line plot — immigration trend 1980–2013 with markers, grid, legend and axis limits.

1 & 2 & 3. Box Plot — Distribution per Country per Year

Box plot of immigrants per country for each year. Shows median, IQR and outliers. Styled with blue fill, red median line.

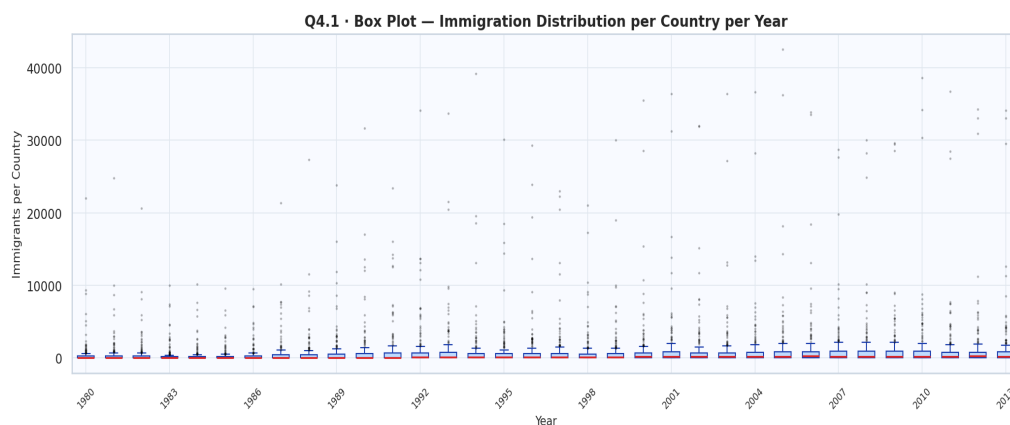


Figure 14: Box plot showing immigration distribution per country across years 1980–2013.

1 & 2 & 3. Scatter Plot — Color-Coded by Volume

Scatter plot with colormap='plasma' to encode immigration volume. Dashed trendline added. Colorbar legend shown.

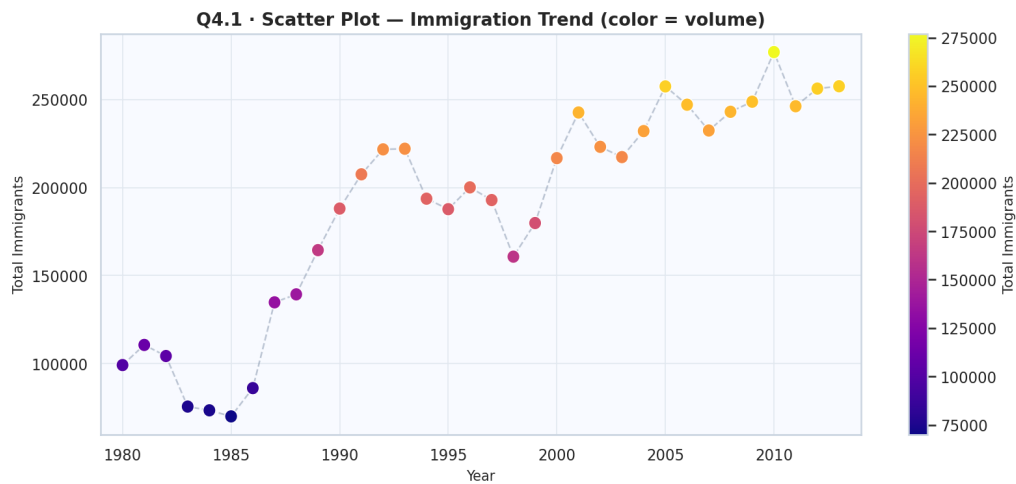


Figure 15: Scatter plot — each point represents one year, colored by total immigration volume.

4. Frequency Distribution — New Immigrants to Canada in 2013

Histogram showing how many countries sent a given number of immigrants in 2013. Most countries sent very few; a small number sent very large numbers (right-skewed). Red dashed line marks the mean.

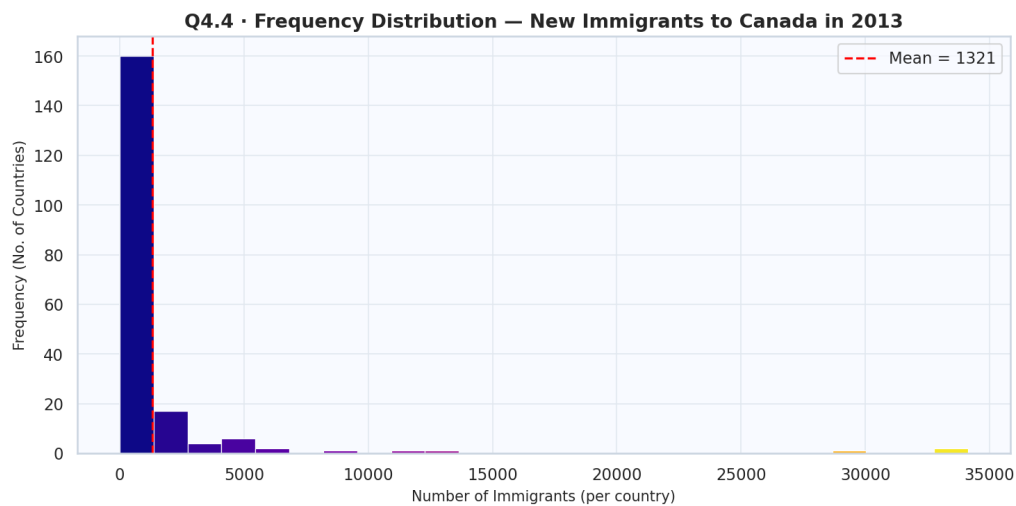


Figure 16: Frequency distribution of immigration counts per country for year 2013.

5. Immigration: Denmark, Norway & Sweden (1980–2013)

Line chart comparing immigration from three Scandinavian countries. Each country uses distinct color, marker and linestyle for clear differentiation.

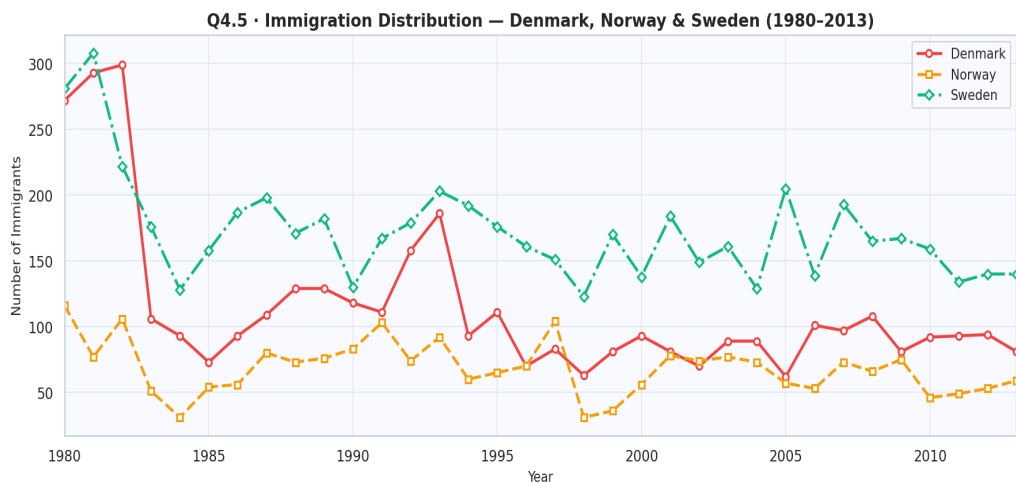


Figure 17: Immigration distribution for Denmark (red), Norway (amber), Sweden (green).

6. Pie Chart — Total Immigrants by Continent

Aggregated 'Total' column by Continent. Asia and Europe dominate. Largest slice exploded for emphasis.

Q4.6 · Total Immigrants Proportion by Continent (1980–2013)

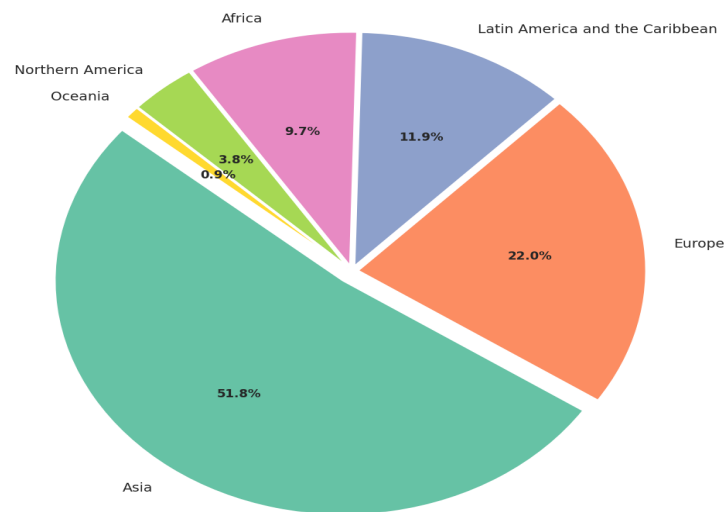


Figure 18: Pie chart — proportion of total immigrants (1980–2013) by continent.

7. Line & Scatter Subplot in One Row

Used `plt.subplots(1, 2)` to place line plot and scatter plot side-by-side in a single figure row. Shared y-axis. Scatter uses 'viridis' colormap.

Q4.7 · Line & Scatter Plot in One Row (Subplots) — Canada Immigration 1980–2013

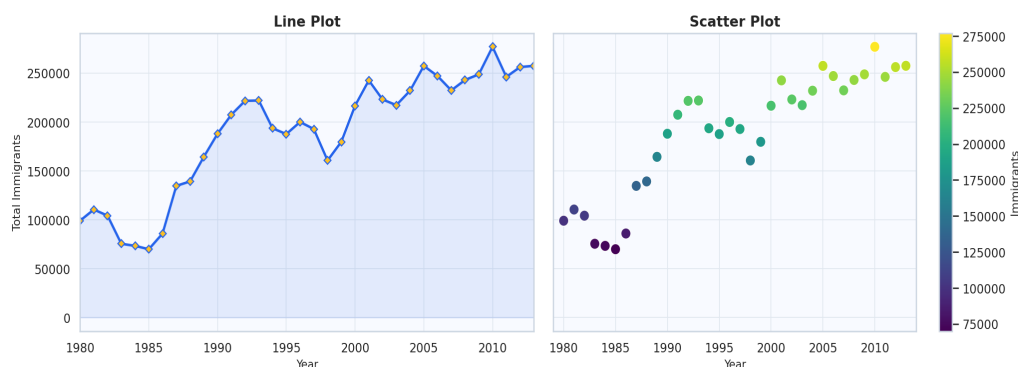


Figure 19: Subplots — line chart (left) and scatter plot (right) in one row.

Python Code — Q4 Key Snippets:

```
total_yr = df_can[years].sum() # sum all countries per year
# Line plot (Q4.1)
ax.plot(years, total_yr, 'D-', color='#2563EB', markersize=5,
        markerfacecolor='#FBBF24', label='Total Immigrants')
ax.grid(True, linestyle='--', alpha=0.5); ax.legend(); ax.set_xlim(1980,2013)
# Freq distribution (Q4.4)
ax.hist(df_can[2013], bins=25, cmap='plasma')
# Pie chart (Q4.6)
cont = df_can.groupby('Continent')['Total'].sum()
ax.pie(cont, labels=cont.index, autopct='%1.1f%%', explode=[0.05,0.02,...])
# Subplots (Q4.7)
fig, (ax1,ax2) = plt.subplots(1,2,figsize=(14,5))
ax1.plot(years, total_yr, 'D-', color='#2563EB')
ax2.scatter(years, total_yr, c=total_yr, cmap='viridis', s=90)
```

Assignment 1 — Completed

All 4 Questions Answered with Code, Analysis & Visualizations

Question	Topic	Dataset	Status
Q 1	Data Wrangling — Cleaning, Filtering & Sorting	Canada.xlsx	Complete ✓
Q 2	Data Preprocessing — Missing, Scaling & Encoding	Laptop.csv	Complete ✓
Q 3	Hypothesis Testing — t-test, Chi ² , ANOVA	Insurance.csv	Complete ✓
Q 4	Visualization — Line, Box, Scatter, Pie, Subplots	Canada.xlsx	Complete ✓

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